

Cyclists Case Study Project

Ask

The goal of this analysis is to understand how annual members and casual riders use Cyclistic bikes differently. These insights can help the marketing team develop strategies to convert casual riders into annual members and support business decisions.

Stakeholders: Cyclistic Marketing Analytics Team, Cyclistic Executive Team, and Lily Moreno (Director of Marketing).

Prepare

In the preparation stage, the data comes from a set of CSV files provided by the Google Data Analytics course, stored in Google Cloud Storage. The data is organized by month, with each file containing ride records, including trip IDs, bike type, start and end times, and stations. While I identified some empty cells, they were deemed irrelevant to the main analysis, so I did not modify the tables. The data integrity was ensured by reviewing key columns, and I assumed the original source was reliable. This consistent and clear data allows me to proceed with the analysis of usage patterns, differentiating between annual members and casual riders, without introducing bias or altering the source.

Process

In this processing phase, I first used Google Sheets to perform some basic calculations and get a better understanding of the data. Once I had a clearer picture, I chose SQL as the main tool for cleaning the data. SQL allowed me to filter, remove duplicates, and calculate new columns, such as ride duration and day of the week. I documented each cleaning step to ensure transparency, and I verified data integrity by checking key columns for consistency. I am planning to use Tableau for the final visualizations.

Analyze

The total number of rides is approximately 5.5 million. Of these, 64% are annual members, while 35% are casual riders. The average ride duration is 22 minutes, but members have a shorter average of 12 minutes. During weekdays, rides account for 72% of total usage, while weekends account for 28%, with weekday average ride times at 14 minutes. The bike usage peaks in the afternoon (46%), followed by morning and night. I also observed a seasonal pattern: rides increased significantly in July and August, likely due to summer, while the first

and last months of the year saw the fewest rides. These insights help Cyclistic better understand usage patterns, guiding decisions on bike availability, marketing efforts, and seasonal strategies.

Share

I was able to answer the question of how annual members and casual riders use Cyclistic bikes differently. The data reveals a clear story: 64% of rides come from members, and their average ride is shorter (12 minutes) compared to casual riders (22 minutes). Weekdays dominate usage (72%), especially in the afternoon (46%), while weekends see fewer rides (28%). I can see seasonal patterns as well, with peaks in July and August. My audience, including the marketing team, benefits from these insights, as data visualization in Tableau helps them quickly grasp trends and adjust strategies. The presentation is accessible, with clear visuals and concise explanations, ensuring the team can act on these findings.

Key Insights:

- **64% of all rides are made by annual members.** Members generate the majority of trips, indicating frequent use of Cyclistic bikes.
- **Casual riders take longer trips.** Their average ride length is almost twice that of members, suggesting more leisure-oriented usage.
- **Bike usage is concentrated on weekdays.** Approximately 72% of rides occur during the workweek, indicating that many users rely on bikes for commuting.
- **Demand peaks during summer months.** Ride activity reaches its highest levels in July and August and declines during colder months.
- **Afternoon is the most active period.** Most rides take place in the afternoon, followed by morning and nighttime usage.

Act

Based on the analysis, my final conclusion is that annual members account for the majority of rides, even though casual riders take a notable share of shorter trips during weekday mornings and afternoons. These insights suggest that we can focus on converting active casual riders into annual members by leveraging their weekday commuting habits. My three recommendations are: first, personalize offers for frequent casual riders who commute during the week, emphasizing the cost savings of membership; second, analyze trip origin and destination data to identify key commuting routes and promote memberships for those corridors; and third, continue gathering data on trip timing and purpose, which will allow us to

refine and personalize marketing strategies even further. These steps will help convert habitual casual users into loyal members, driving sustainable growth.